stick to imprint signs on a piece of wet clay. Could we, at that moment, envision the Library of Alexandria, the power of the Bible or the archives of the NKVD? Even this analogy grossly underestimates the potential of future computer evolution. So try to imagine that we are now eighty years since the first self-replicating genetic code lines coalesced out of the organic soup of early Earth, about four billion years ago. At this stage, even single-celled amoebas with their cellular organisation, their thousands of internal organelles and their ability to control movement and nutrition are still futuristic fantasies.[45] Could we envision *Tyrannosaurus rex*, the Amazon rain forest, or humans landing on the moon?

We still tend to think of a computer as a metal box with a screen and a keyboard, because this is the shape our organic imagination gave to the baby computers in the twentieth century. As computers grow and develop, they are shedding old forms and taking radically new configurations, breaking the limits of human imagination. Unlike organic beings, computers don't have to be in just one place at any one time. They diffuse over space, with parts in different cities and continents. In computer evolution, the distance from amoeba to *T. rex* could be covered in a decade. If GPT-4 is the amoeba, what would the *T. rex* look like? Organic evolution took four billion years to get from organic soup to apes on the moon. Computers may require just a few centuries to develop superintelligence, expand to planetary sizes, contract to a subatomic level or sprawl over galactic space and time.

The pace of computer evolution is reflected in the terminological chaos that surrounds computers. While a couple of decades ago it was customary to speak only about 'computers', now we find ourselves talking about algorithms, robots, bots, AIs, networks or clouds. Our difficulty in deciding what to call them is itself important. Organisms are distinct individual entities that can be grouped into collectives like species and genera. With computers, however, it is becoming ever more difficult to decide where one entity ends and another begins and how exactly to group them.

In this book I use the term 'computer' when talking about the whole complex of software and hardware, manifested in physical form. I prefer to often use the almost-archaic-sounding 'computer' over 'algorithm' or 'AI', partly because I am aware of how fast terms change and partly to remind us of the physical aspect of the computer revolution. Computers are made of matter, they consume energy and they fill a space. Enormous amounts of